EXPERIMENT -3

AIM- Create an Employee–Department–Project schema. Insert at least 30 employees across 5 departments and 8 projects. Write SQL queries demonstrating selection, projection, aggregates, GROUP BY, HAVING, CASE expressions, and ORDER BY.

OBJECTIVES

- To create relational tables for Department, Employee, and Project.
- To establish relationships between the tables using Primary Keys and Foreign Keys.
- To insert at least 30 employees, 5 departments, and 8 projects.
- To retrieve specific records using Selection.
- To retrieve specific attributes using Projection.
- To perform calculations using aggregate functions such as COUNT(), AVG(), MAX(), MIN(), and SUM().
- To group records using GROUP BY.
- To filter grouped results using HAVING.
- To categorize employee data using a CASE expression. ● To sort query results using ORDER BY.

THEORY

DATABASE SCHEMA

A database schema defines the logical structure of a database, including its tables, attributes, and relationships. In this experiment, the database consists of the Department, Employee, Project, and Employee_Project relations. The Department relation stores department information, the Employee relation stores employee details, the Project relation stores project information, and the Employee_Project relation associates employees with projects. The database contains 5 departments, 8 projects, and 30 employees.

SELECTION

Selection is used to retrieve records from a relation that satisfy a specified condition. In SQL, selection is generally performed using the WHERE clause. It filters the rows of a table according to the given condition.

PROJECTION

Projection is used to retrieve only the required attributes or columns from a relation. In SQL, projection is performed using the SELECT statement by specifying the required columns. It allows only the relevant attributes to be displayed in the result.

AGGREGATE FUNCTIONS

Aggregate functions perform calculations on a set of rows and return a single result. The aggregate functions used in this experiment are:

- COUNT() – Counts the number of rows.
- SUM() – Calculates the total of numeric values.
- AVG() – Calculates the average of numeric values.
- MAX() – Returns the maximum value.
- MIN() – Returns the minimum value.

GROUP BY

The GROUP BY clause is used to arrange rows into groups based on one or more columns. It is commonly used with aggregate functions to perform calculations separately for each group. In this experiment, it is used for department-wise analysis of employees and salaries.

HAVING

The HAVING clause is used to apply conditions to groups created using the GROUP BY clause. It is generally used with aggregate functions to filter grouped results based on a specified condition.

CASE EXPRESSION

The CASE expression is used to perform conditional operations in SQL. It evaluates specified conditions and returns a corresponding value when a condition is satisfied. In this experiment, it is used to categorize employees based on their salary.

ORDER BY

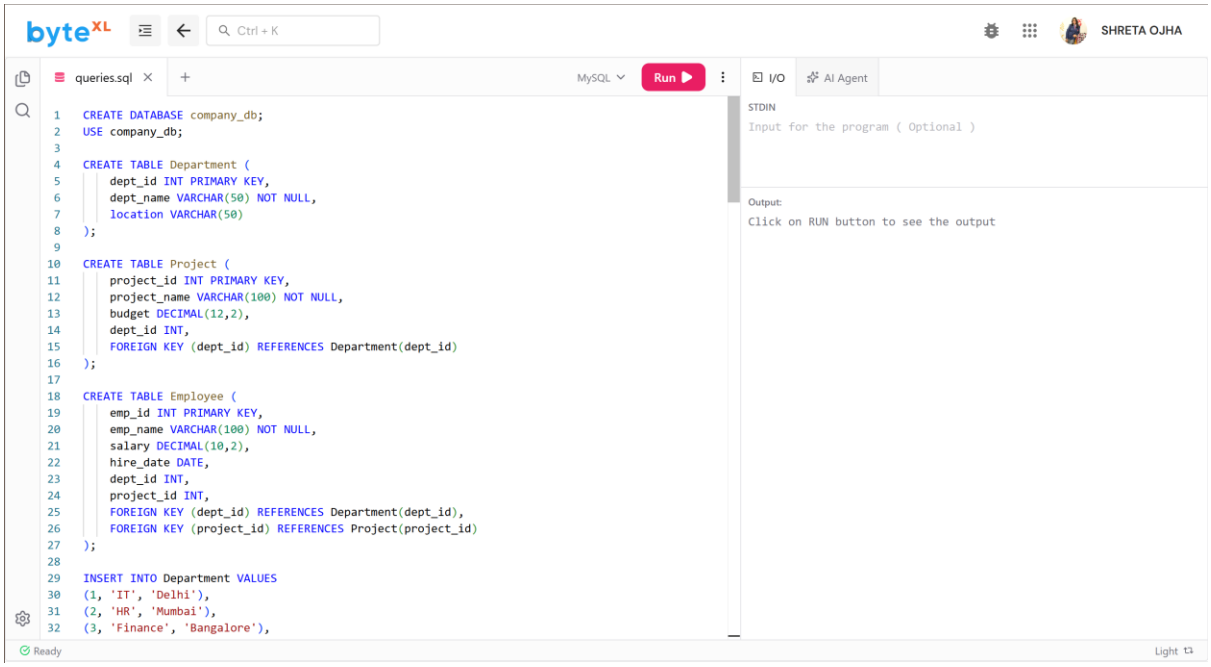
The ORDER BY clause is used to sort the records returned by a query. The results can be arranged in ascending order using ASC or descending order using DESC. In this experiment, it is used to arrange employees according to their salary.

DATABASE RELATIONSHIPS

The database represents the relationship between employees, departments, and projects. A department can have multiple employees and projects. An employee can work on

multiple projects, and a project can have multiple employees. The Employee_Project relation is used to represent the association between employees and projects.

SOURCE CODE-



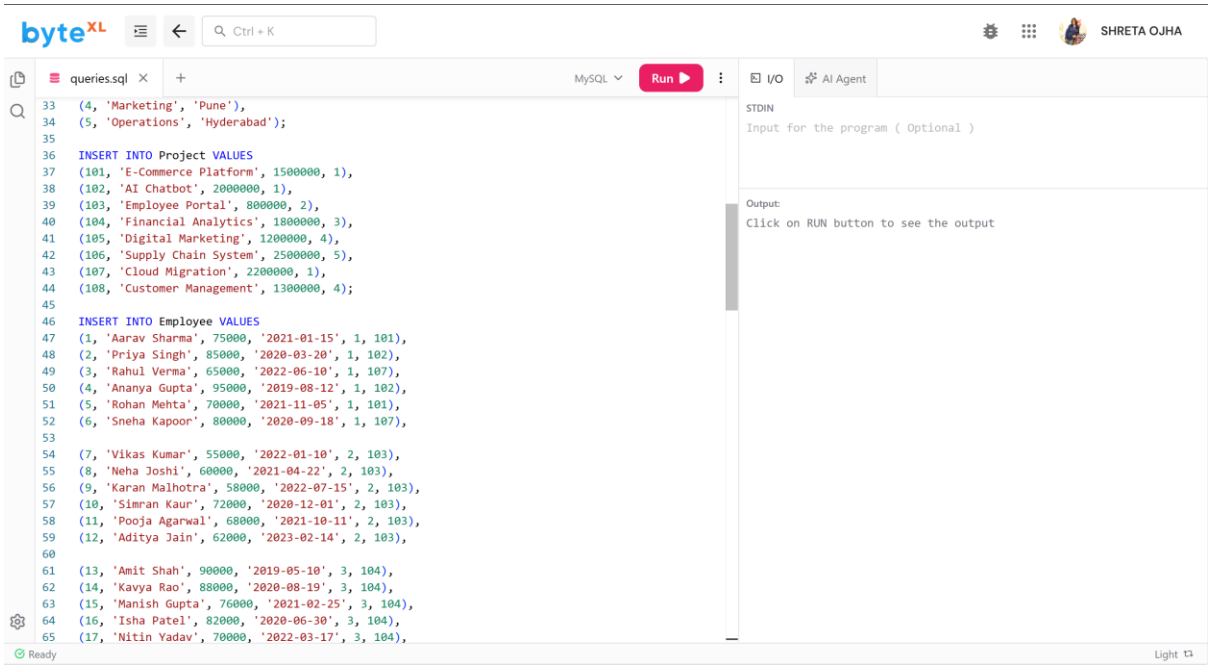
```sql
CREATE DATABASE company_db;
USE company_db;

CREATE TABLE Department (
    dept_id INT PRIMARY KEY,
    dept_name VARCHAR(50) NOT NULL,
    location VARCHAR(50)
);

CREATE TABLE Project (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100) NOT NULL,
    budget DECIMAL(12,2),
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(100) NOT NULL,
    salary DECIMAL(10,2),
    hire_date DATE,
    dept_id INT,
    project_id INT,
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
    FOREIGN KEY (project_id) REFERENCES Project(project_id)
);

INSERT INTO Department VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Mumbai'),
(3, 'Finance', 'Bangalore'),
```



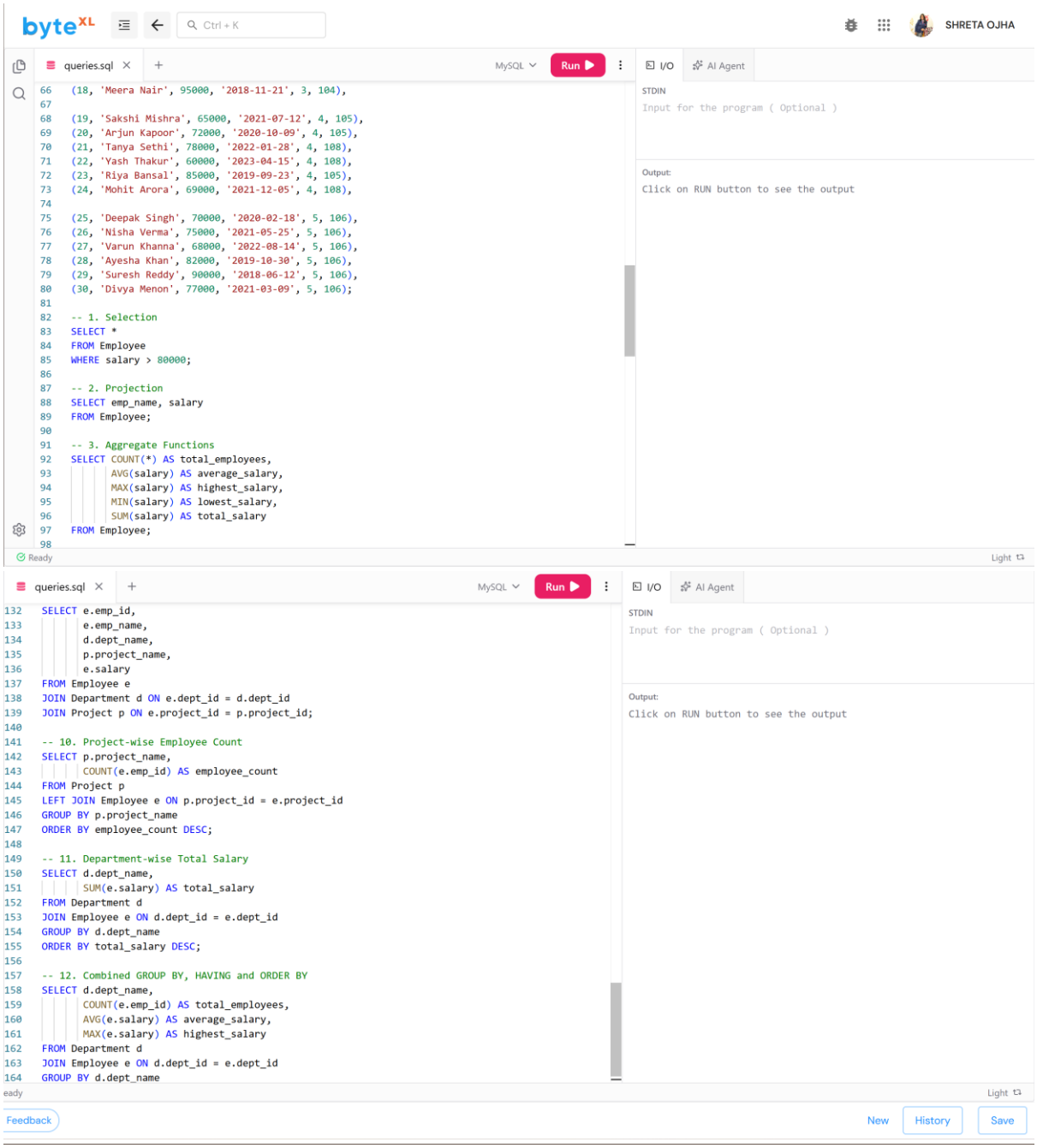
```sql
(4, 'Marketing', 'Pune'),
(5, 'Operations', 'Hyderabad');

INSERT INTO Project VALUES
(101, 'E-Commerce Platform', 1500000, 1),
(102, 'AI Chatbot', 2000000, 1),
(103, 'Employee Portal', 800000, 2),
(104, 'Financial Analytics', 1800000, 3),
(105, 'Digital Marketing', 1200000, 4),
(106, 'Supply Chain System', 2500000, 5),
(107, 'Cloud Migration', 2200000, 1),
(108, 'Customer Management', 1300000, 4);

INSERT INTO Employee VALUES
(1, 'Aarav Sharma', 75000, '2021-01-15', 1, 101),
(2, 'Priya Singh', 85000, '2020-03-20', 1, 102),
(3, 'Rahul Verma', 65000, '2022-06-10', 1, 107),
(4, 'Ananya Gupta', 95000, '2019-08-12', 1, 102),
(5, 'Rohan Mehta', 70000, '2021-11-05', 1, 101),
(6, 'Sneha Kapoor', 80000, '2020-09-18', 1, 107),

(7, 'Vikas Kumar', 55000, '2022-01-10', 2, 103),
(8, 'Neha Joshi', 60000, '2021-04-22', 2, 103),
(9, 'Karan Malhotra', 58000, '2022-07-15', 2, 103),
(10, 'Simran Kaur', 72000, '2020-12-01', 2, 103),
(11, 'Pooja Agarwal', 68000, '2021-10-11', 2, 103),
(12, 'Aditya Jain', 62000, '2023-02-14', 2, 103),

(13, 'Amit Shah', 90000, '2019-05-10', 3, 104),
(14, 'Kavya Rao', 88000, '2020-08-19', 3, 104),
(15, 'Manish Gupta', 76000, '2021-02-25', 3, 104),
(16, 'Isha Patel', 82000, '2020-06-30', 3, 104),
(17, 'Nitin Yadav', 70000, '2022-03-17', 3, 104),
```




```
(18, 'Meera Nair', 95000, '2018-11-21', 3, 104),

(19, 'Sakshi Mishra', 65000, '2021-07-12', 4, 105),
(20, 'Arjun Kapoor', 72000, '2020-10-09', 4, 105),
(21, 'Tanya Sethi', 78000, '2022-01-28', 4, 108),
(22, 'Yash Thakur', 60000, '2023-04-15', 4, 108),
(23, 'Riya Bansal', 85000, '2019-09-23', 4, 105),
(24, 'Mohit Arora', 69000, '2021-12-05', 4, 108),

(25, 'Deepak Singh', 70000, '2020-02-18', 5, 106),
(26, 'Nisha Verma', 75000, '2021-05-25', 5, 106),
(27, 'Varun Khanna', 68000, '2022-08-14', 5, 106),
(28, 'Ayesha Khan', 82000, '2019-10-30', 5, 106),
(29, 'Suresh Reddy', 90000, '2018-06-12', 5, 106),
(30, 'Divya Menon', 77000, '2021-03-09', 5, 106);

-- 1. Selection
SELECT *
FROM Employee
WHERE salary > 80000;

-- 2. Projection
SELECT emp_name, salary
FROM Employee;

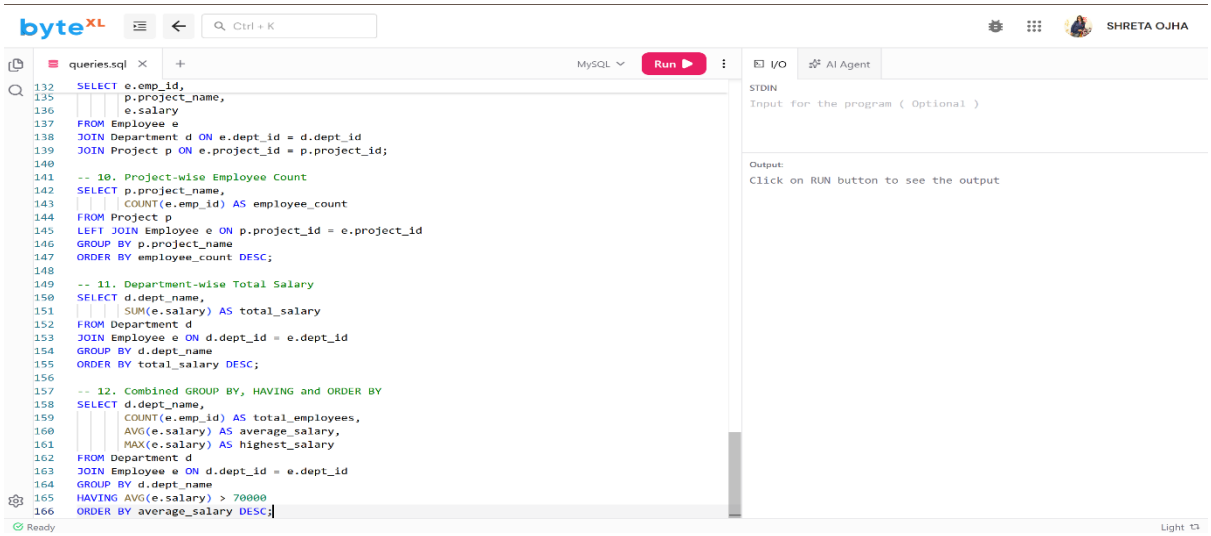
-- 3. Aggregate Functions
SELECT COUNT(*) AS total_employees,
       AVG(salary) AS average_salary,
       MAX(salary) AS highest_salary,
       MIN(salary) AS lowest_salary,
       SUM(salary) AS total_salary
FROM Employee;
```

```
SELECT e.emp_id,
       e.emp_name,
       d.dept_name,
       p.project_name,
       e.salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
JOIN Project p ON e.project_id = p.project_id;

-- 10. Project-wise Employee Count
SELECT p.project_name,
       COUNT(e.emp_id) AS employee_count
FROM Project p
LEFT JOIN Employee e ON p.project_id = e.project_id
GROUP BY p.project_name
ORDER BY employee_count DESC;

-- 11. Department-wise Total Salary
SELECT d.dept_name,
       SUM(e.salary) AS total_salary
FROM Department d
JOIN Employee e ON d.dept_id = e.dept_id
GROUP BY d.dept_name
ORDER BY total_salary DESC;

-- 12. Combined GROUP BY, HAVING and ORDER BY
SELECT d.dept_name,
       COUNT(e.emp_id) AS total_employees,
       AVG(e.salary) AS average_salary,
       MAX(e.salary) AS highest_salary
FROM Department d
JOIN Employee e ON d.dept_id = e.dept_id
GROUP BY d.dept_name
```



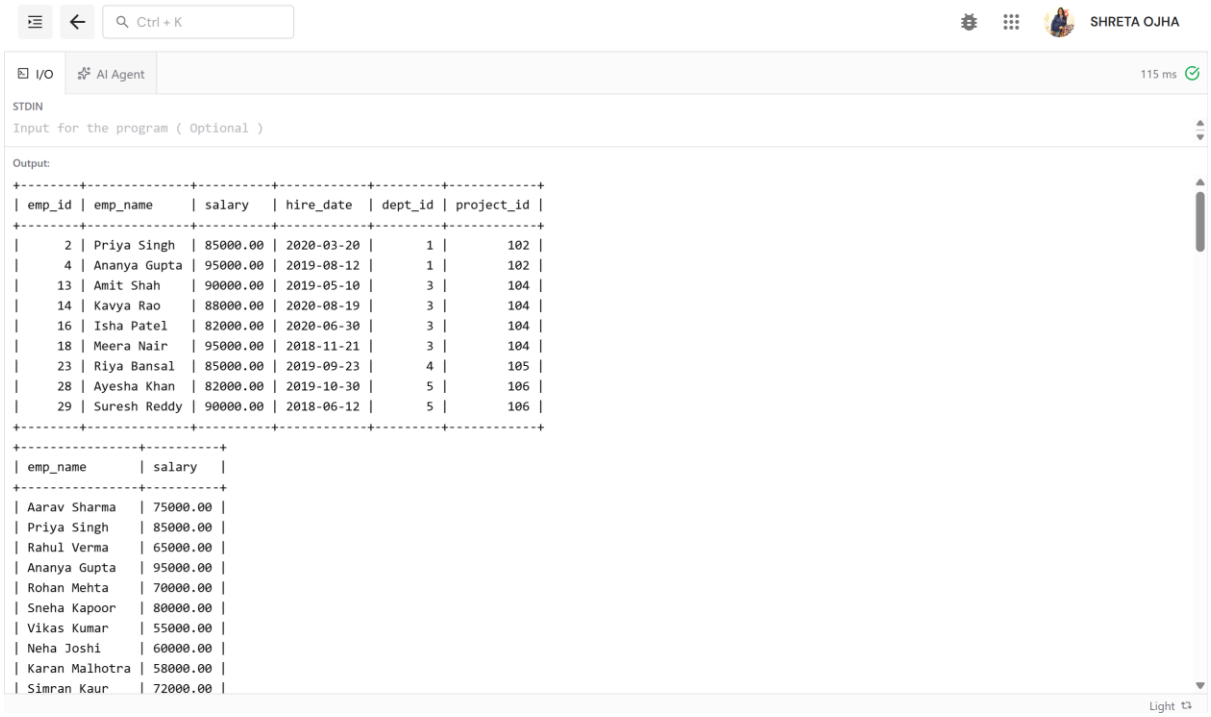

```
SELECT e.emp_id,
       p.project_name,
       e.salary
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id
JOIN Project p ON e.project_id = p.project_id;

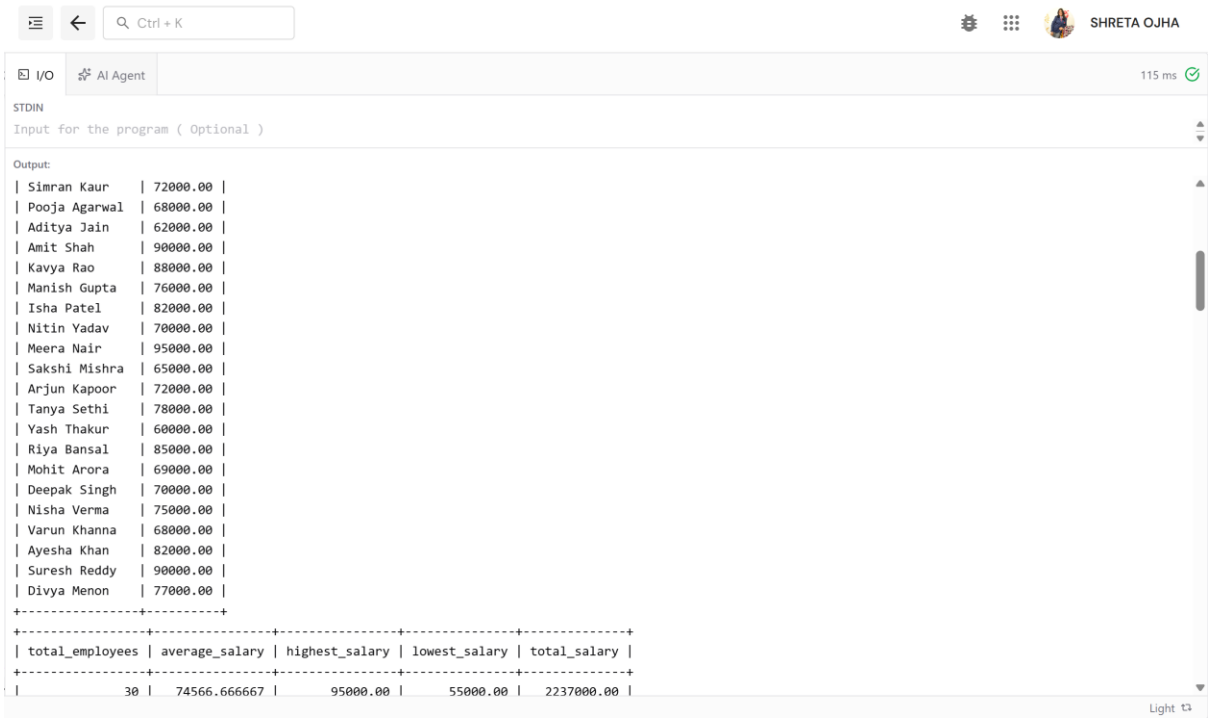
-- 10. Project-wise Employee Count
SELECT p.project_name,
       COUNT(e.emp_id) AS employee_count
FROM Project p
LEFT JOIN Employee e ON p.project_id = e.project_id
GROUP BY p.project_name
ORDER BY employee_count DESC;

-- 11. Department-wise Total Salary
SELECT d.dept_name,
       SUM(e.salary) AS total_salary
FROM Department d
JOIN Employee e ON d.dept_id = e.dept_id
GROUP BY d.dept_name
ORDER BY total_salary DESC;

-- 12. Combined GROUP BY, HAVING and ORDER BY
SELECT d.dept_name,
       COUNT(e.emp_id) AS total_employees,
       AVG(e.salary) AS average_salary,
       MAX(e.salary) AS highest_salary
FROM Department d
JOIN Employee e ON d.dept_id = e.dept_id
GROUP BY d.dept_name
HAVING AVG(e.salary) > 70000
ORDER BY average_salary DESC;
```

OUTPUT


Ctrl + K
SHRETA OJHA
I/O
AI Agent
115 ms
STDIN
Input for the program (Optional)
Output:
+--------+--------------+----------+------------+---------+------------+
| emp_id | emp_name | salary | hire_date | dept_id | project_id |
+--------+--------------+----------+------------+---------+------------+
2	Priya Singh	85000.00	2020-03-20	1	102
4	Ananya Gupta	95000.00	2019-08-12	1	102
13	Amit Shah	90000.00	2019-05-10	3	104
14	Kavya Rao	88000.00	2020-08-19	3	104
16	Isha Patel	82000.00	2020-06-30	3	104
18	Meera Nair	95000.00	2018-11-21	3	104
23	Riya Bansal	85000.00	2019-09-23	4	105
28	Ayesha Khan	82000.00	2019-10-30	5	106
29	Suresh Reddy	90000.00	2018-06-12	5	106
+--------+--------------+----------+------------+---------+------------+
+----------------+----------+
| emp_name | salary |
+----------------+----------+
Aarav Sharma	75000.00
Priya Singh	85000.00
Rahul Verma	65000.00
Ananya Gupta	95000.00
Rohan Mehta	70000.00
Sneha Kapoor	80000.00
Vikas Kumar	55000.00
Neha Joshi	60000.00
Karan Malhotra	58000.00
Simran Kaur	72000.00
Light


Ctrl + K
SHRETA OJHA
I/O
AI Agent
115 ms
STDIN
Input for the program (Optional)
Output:
Simran Kaur	72000.00
Pooja Agarwal	68000.00
Aditya Jain	62000.00
Amit Shah	90000.00
Kavya Rao	88000.00
Manish Gupta	76000.00
Isha Patel	82000.00
Nitin Yadav	70000.00
Meera Nair	95000.00
Sakshi Mishra	65000.00
Arjun Kapoor	72000.00
Tanya Sethi	78000.00
Yash Thakur	60000.00
Riya Bansal	85000.00
Mohit Arora	69000.00
Deepak Singh	70000.00
Nisha Verma	75000.00
Varun Khanna	68000.00
Ayesha Khan	82000.00
Suresh Reddy	90000.00
Divya Menon	77000.00
+----------------+----------+
+-----------------+-----------------+----------------+---------------+--------------+
| total_employees | average_salary | highest_salary | lowest_salary | total_salary |
+-----------------+-----------------+----------------+---------------+--------------+
| 30 | 74566.666667 | 95000.00 | 55000.00 | 2237000.00 |
Light

I/O | AI Agent | 115 ms

STDIN

Input for the program (Optional)

Output:

```
+----------------+----------------+----------------+---------------+--------------+
| total_employees | average_salary | highest_salary | lowest_salary | total_salary |
+----------------+----------------+----------------+---------------+--------------+
|              30 |   74566.666667 |       95000.00 |      55000.00 |   2237000.00 |
+----------------+----------------+----------------+---------------+--------------+
+---------+----------------+
| dept_id | employee_count |
+---------+----------------+
|       1 |              6 |
|       2 |              6 |
|       3 |              6 |
|       4 |              6 |
|       5 |              6 |
+---------+----------------+
+------------+----------------+
| dept_name  | employee_count |
+------------+----------------+
| IT         |              6 |
| HR         |              6 |
| Finance    |              6 |
| Marketing  |              6 |
| Operations |              6 |
+------------+----------------+
+---------+----------------+
| dept_id | average_salary |
+---------+----------------+
```

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SHRETA OJHA

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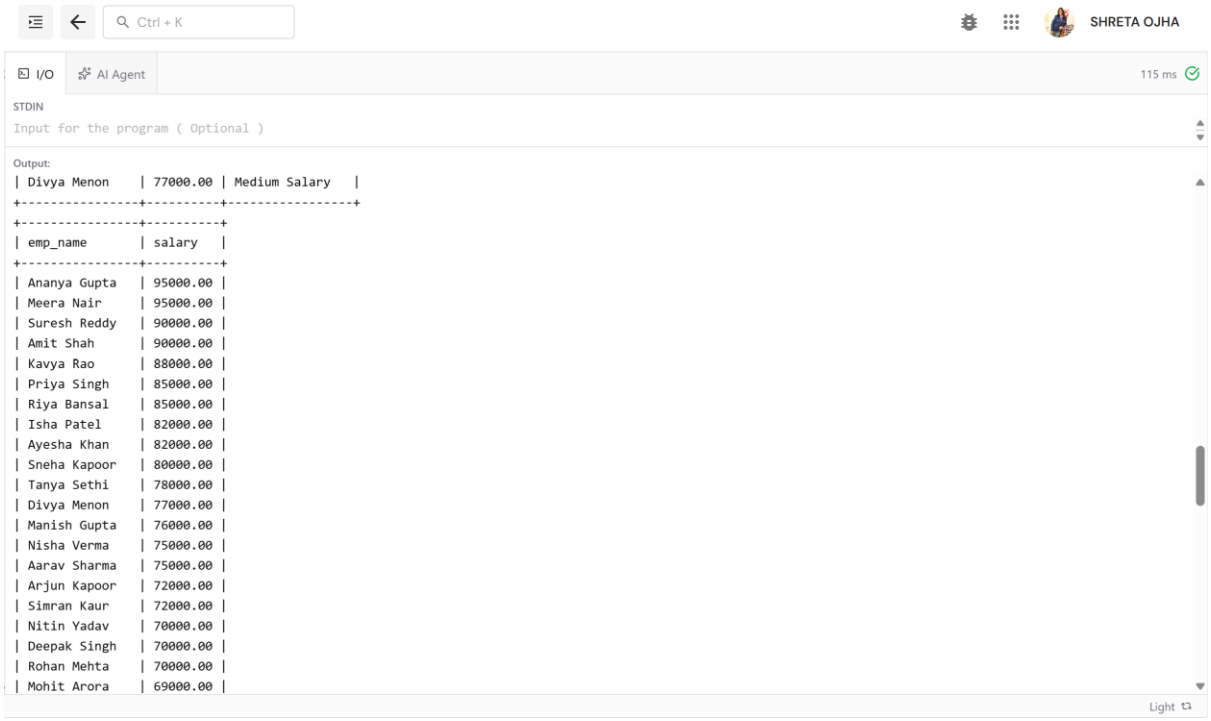
STDIN

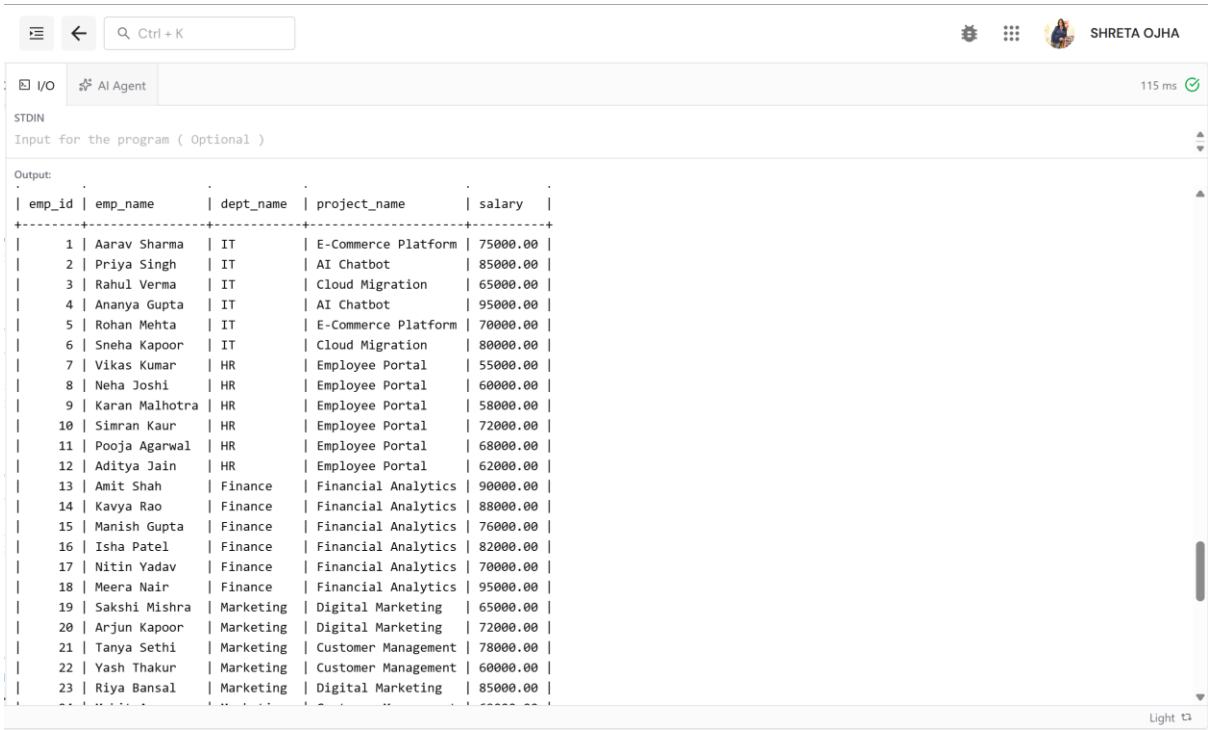
Input for the program (Optional)

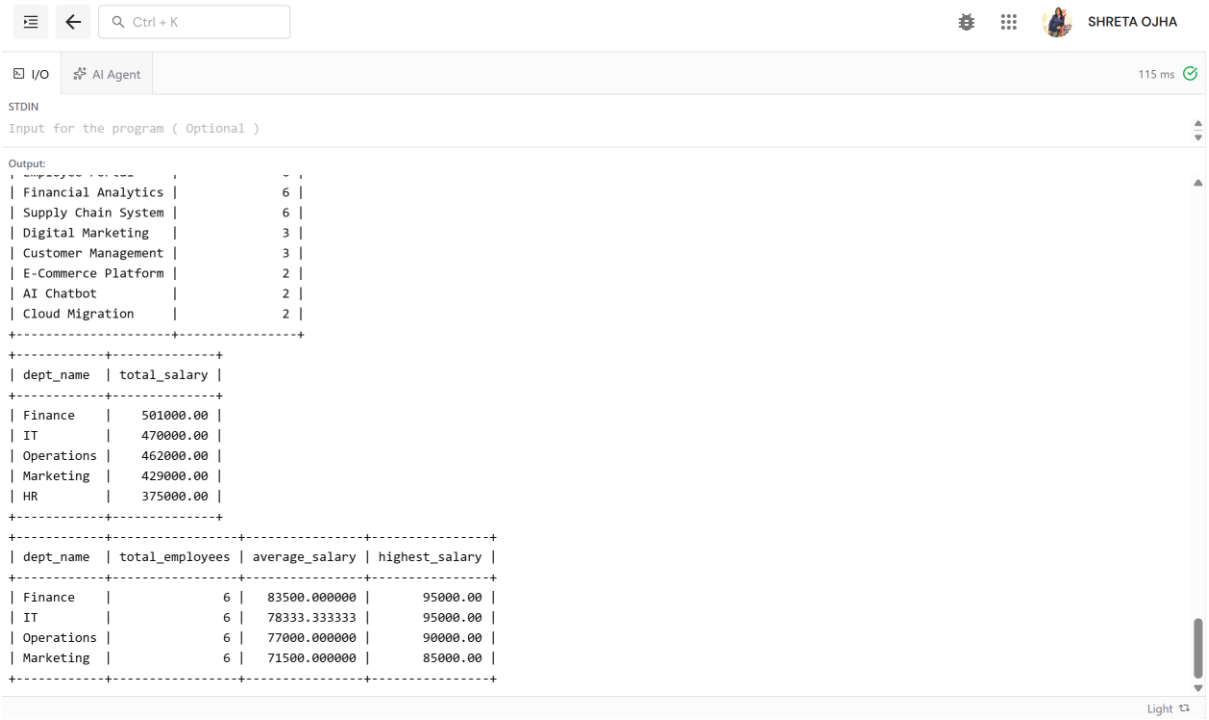
Output:

```
+---------+----------------+
| dept_id | average_salary |
+---------+----------------+
|       1 |   78333.333333 |
|       3 |   83500.000000 |
|       5 |   77000.000000 |
+---------+----------------+
+---------------+----------+-----------------+
| emp_name      | salary   | salary_category |
+---------------+----------+-----------------+
| Aarav Sharma  | 75000.00 | Medium Salary   |
| Priya Singh   | 85000.00 | Medium Salary   |
| Rahul Verma   | 65000.00 | Low Salary      |
| Ananya Gupta  | 95000.00 | High Salary     |
| Rohan Mehta   | 70000.00 | Medium Salary   |
| Sneha Kapoor  | 80000.00 | Medium Salary   |
| Vikas Kumar   | 55000.00 | Low Salary      |
| Neha Joshi    | 60000.00 | Low Salary      |
| Karan Malhotra | 58000.00 | Low Salary      |
| Simran Kaur   | 72000.00 | Medium Salary   |
| Pooja Agarwal | 68000.00 | Low Salary      |
| Aditya Jain   | 62000.00 | Low Salary      |
| Amit Shah     | 90000.00 | High Salary     |
| Kavya Rao     | 88000.00 | Medium Salary   |
| Manish Gupta  | 76000.00 | Medium Salary   |
| Isha Patel    | 82000.00 | Medium Salary   |
```

Light


Ctrl + K
SHRETA OJHA
I/O
AI Agent
115 ms
STDIN
Input for the program (Optional)
Output:
| Divya Menon | 77000.00 | Medium Salary |
+---------------+----------+-----------------+
+---------------+----------+
| emp_name | salary |
+---------------+----------+
Ananya Gupta	95000.00
Meera Nair	95000.00
Suresh Reddy	90000.00
Amit Shah	90000.00
Kavya Rao	88000.00
Priya Singh	85000.00
Riya Bansal	85000.00
Isha Patel	82000.00
Ayesha Khan	82000.00
Sneha Kapoor	80000.00
Tanya Sethi	78000.00
Divya Menon	77000.00
Manish Gupta	76000.00
Nisha Verma	75000.00
Aarav Sharma	75000.00
Arjun Kapoor	72000.00
Simran Kaur	72000.00
Nitin Yadav	70000.00
Deepak Singh	70000.00
Rohan Mehta	70000.00
Mohit Arora	69000.00
Light


Ctrl + K
SHRETA OJHA
I/O
AI Agent
115 ms
STDIN
Input for the program (Optional)
Output:
emp_id	emp_name	dept_name	project_name	salary
1	Aarav Sharma	IT	E-Commerce Platform	75000.00
2	Priya Singh	IT	AI Chatbot	85000.00
3	Rahul Verma	IT	Cloud Migration	65000.00
4	Ananya Gupta	IT	AI Chatbot	95000.00
5	Rohan Mehta	IT	E-Commerce Platform	70000.00
6	Sneha Kapoor	IT	Cloud Migration	80000.00
7	Vikas Kumar	HR	Employee Portal	55000.00
8	Neha Joshi	HR	Employee Portal	60000.00
9	Karan Malhotra	HR	Employee Portal	58000.00
10	Simran Kaur	HR	Employee Portal	72000.00
11	Pooja Agarwal	HR	Employee Portal	68000.00
12	Aditya Jain	HR	Employee Portal	62000.00
13	Amit Shah	Finance	Financial Analytics	90000.00
14	Kavya Rao	Finance	Financial Analytics	88000.00
15	Manish Gupta	Finance	Financial Analytics	76000.00
16	Isha Patel	Finance	Financial Analytics	82000.00
17	Nitin Yadav	Finance	Financial Analytics	70000.00
18	Meera Nair	Finance	Financial Analytics	95000.00
19	Sakshi Mishra	Marketing	Digital Marketing	65000.00
20	Arjun Kapoor	Marketing	Digital Marketing	72000.00
21	Tanya Sethi	Marketing	Customer Management	78000.00
22	Yash Thakur	Marketing	Customer Management	60000.00
23	Riya Bansal	Marketing	Digital Marketing	85000.00
Light



OBSERVATION/OUTCOME

1. Employee, Department, and Project tables were successfully created.
2. Data for 30 employees, 5 departments, and 8 projects was successfully inserted.
3. Selection and projection queries successfully retrieved the required records and attributes.
4. Aggregate functions such as COUNT(), SUM(), AVG(), MAX(), and MIN() were successfully executed.
5. GROUP BY and HAVING were used to group and filter grouped data. 6. CASE expression successfully categorized employees based on salary.
6. ORDER BY successfully sorted employee records according to the specified conditions.
7. Queries using joins successfully retrieved related data from multiple tables.